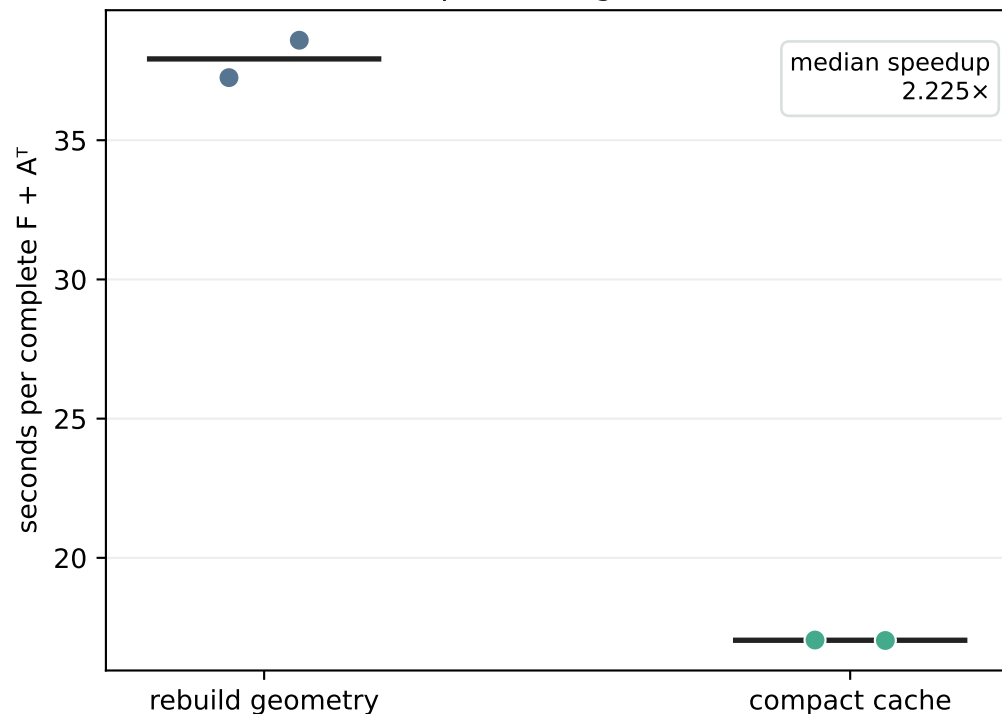
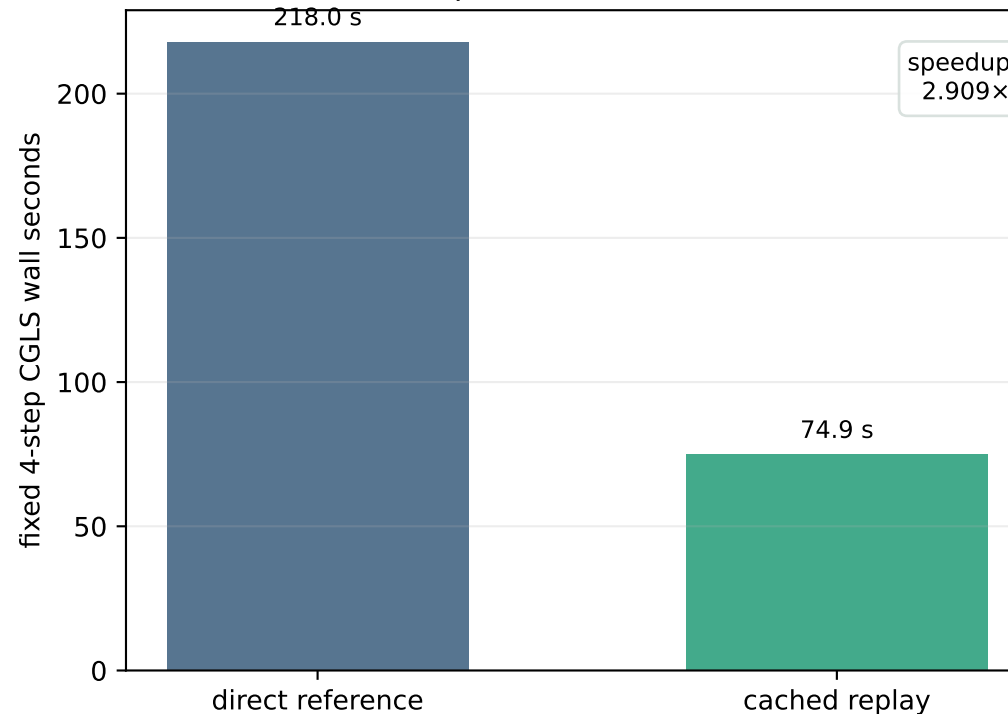


PSU B0 compact stencil cache: exact replay with lower operator cost

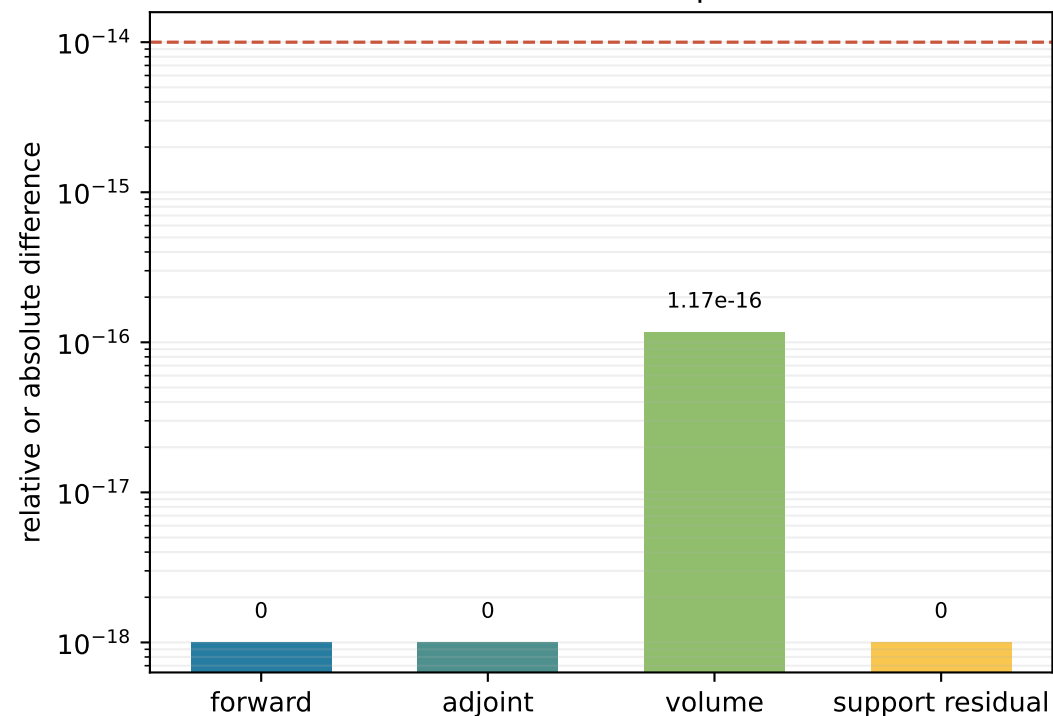
A Same-session full-pair timing



B Frozen 4F + 5A^T optimization



C Direct and cached numerical equivalence



D What the cache changes

10,628,822

support rays

16 per ray

QMC aperture samples

5.017 GB

private cache

14.94 s

one-time cache build

0.627132

same support residual

Execution is faster; the discrete operator, call budget, support fit and evidence level are unchanged.
Rotation-40 development and final audit remain sealed. No experimental 3D ground truth.